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SHORT COMMUNICATION

An analytical solution for temperature field around a cylindrical surface subjected to a time dependent heat flux: An alternative approach



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Abstract An analytical solution for the temperature field in an infinite solid medium around a cylindrical surface subject to a time dependent heat flux proposed by Zanchini and Pulvirenti (2013) has been revisited. The well-known Laplace transform technique is applied to solve the time-dependent governing equation exactly in Laplace domain, while the method of Riemann-sum approximation approach is employed to invert the Laplace domain to the time domain in order to obtain the dimensionless temperature. For the case of sinusoidally varying heat flux, numerical values of the dimensionless temperature of the surface are obtained and compared with the benchmark values reported by Zanchini and Pulvirenti (2013), and an excellent agreement is found. An advantage of the proposed method could be a reduction in computation effort.

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1. Introduction

The problems of heat conduction with time-dependent boundary conditions are of great importance in engineering applications, such as heat exchangers, nuclear reactors, furnaces, surface heat treatments and MEMS (microelectromechanical) devices. The analytical solutions of such problems are used for checking the accuracy of numerical methods of solutions, which can extend engineering utility and give the idea on the stability of the design at large time.

The great importance of time-dependent boundary condition has made a lot of researchers to contribute to this area: Lee and Huang [1] proposed analytical solutions for transient heat conductions in functionally graded slab and in hollow cylinder [2] by using shifting function method. Lee and Te [3] improved [1] by introducing different types of time-dependent boundary conditions and solving the problem by shifting function method developed by Lee and Huang [1].

Zanchini and Pulvirenti [4] obtained an analytical solution for the temperature field around a cylindrical surface subjected to a time dependent heat flux as an extension of Carslaw and Jaeger [5], which holds in case of uniform and constant heat flux. In [4], a general analytical solution was obtained by using the well-known Laplace transform method and the Laplace transform inversion through the convolution integral. Then a special form of the solution, for the case of sinusoidally varying heat flux, was used to determine benchmark values of the

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Nomenclature*List of symbols*

F_0	Fourier number $\left(\frac{\alpha \tau_0}{r_0^2}\right)$
q_0	constant heat flux
r_0	radius of the cylinder
s	Laplace parameter ($s > 0$)
T	dimensional temperature
T_0	initial temperature field within the solid

Greek symbols

k	thermal conductivity
α	thermal diffusivity
η	dimensionless radial coordinate $\left(\eta = \frac{r}{r_0}\right)$
ξ	dimensionless time $\left(\xi = \frac{\tau}{\tau_0}\right)$
τ	dimensional time
τ_0	period
θ	dimensionless temperature $\left(\theta = k \frac{T(r, \tau) - T_0}{q_0 r_0}\right)$

dimensionless temperature of the surface by means of numerical integrations performed through the software Mathematica.

The objective of the present work was to present an alternative solution method to the same problem discussed by [4]. The numerical calculation shows that the present solution predicts the same temperature profiles as those in the published work [4].

2. Mathematical analysis

Reconsider the model proposed by Zanchini and Pulvirenti [4] and incorporating Fig. 1 for more understanding. Following their model and assumptions, the mathematical model for the heat equation in dimensionless form can be written as [4]:

$$\frac{\partial^2 \theta}{\partial \eta^2} + \frac{1}{\eta} \frac{\partial \theta}{\partial \eta} = \frac{\partial \theta}{\partial \xi} \quad (1)$$

$$\theta(\eta, 0) = 0 \quad (2)$$

$$\xi > 0 \begin{cases} \frac{\partial \theta}{\partial \eta} = -\sin\left(\frac{2\pi\xi}{Fo}\right) & \text{at } \eta = 1 \\ \theta(\eta, \xi) \rightarrow \text{finite} & \text{as } \eta \rightarrow \infty \end{cases} \quad (3)$$

The dimensionless quantities used in the above equations are exactly the same as discussed in Zanchini and Pulvirenti [4]. Applying Laplace transform to Eq. (1), $\theta(\eta, \xi)$ with respect to ξ ,

$$\tilde{\theta}(\eta, s) = \int_0^\infty e^{-s\xi} \theta(\eta, \xi) d\xi \quad (4)$$

where $s > 0$ is the Laplace parameter.

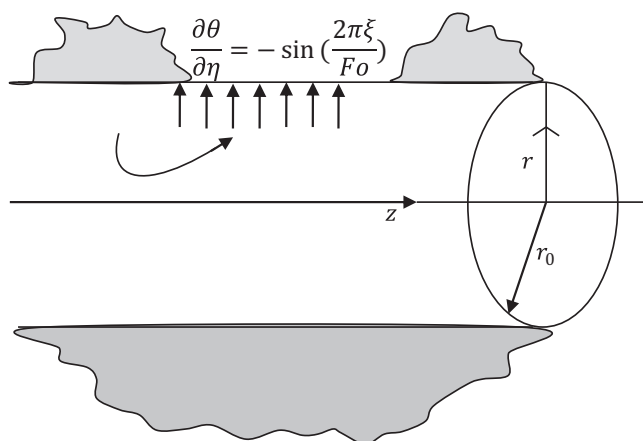


Fig. 1 Schematic diagram of the problem.

In the Laplace transform domain, Eq. (1) with initial and boundary conditions (2) and (3) becomes

$$\frac{\partial^2 \tilde{\theta}}{\partial \eta^2} + \frac{1}{\eta} \frac{\partial \tilde{\theta}}{\partial \eta} - s \tilde{\theta} = 0 \quad (5)$$

$$\frac{\partial \tilde{\theta}(\eta, s)}{\partial \eta} = -\frac{2\pi F_0}{F_0^2 s^2 + 4\pi^2}, \quad \text{at } \eta = 1 \quad (6)$$

$$\tilde{\theta}(\eta, s) \rightarrow \text{finite} \quad \text{as } \eta \rightarrow \infty \quad (7)$$

The solution to Eq. (5) with conditions (6) and (7) is

$$\tilde{\theta}(\eta, s) = \frac{2\pi F_0 K_0(\eta \sqrt{s})}{(F_0^2 s^2 + 4\pi^2) \sqrt{s} K_1(\sqrt{s})} \quad (8)$$

where K_0 and K_1 are modified Bessel's function of second kind of order 0 and 1 respectively.

We used Riemann-sum approximation to find the Laplace inverse of Eq. (8) as discussed in [6–8]. In this method, any function in the Laplace domain can be inverted to time domain as follows:

$$\theta(\eta, \xi) = \frac{e^{\varepsilon \xi}}{\xi} \left[\frac{1}{2} \tilde{\theta}(\eta, \varepsilon) + \text{Re} \sum_{n=1}^M \tilde{\theta} \left(\eta, \varepsilon + \frac{i n \pi}{\xi} \right) (-1)^n \right] \quad (9)$$

where Re refers to the real part, $i = \sqrt{-1}$ called the imaginary number. M is the number of terms used in the Riemann-sum approximation and ε is the real part of the Bromwich contour used in Laplace inversions. The Riemann-sum approximation for Laplace inversion involves a single summation for the numerical process; its accuracy depends on the value of ε and the truncation error is dictated by the number of iterations. According to Tzou [9] the value of $\varepsilon \xi$ that best satisfied the result is 4.7. For faster convergence, however, numerous numerical experiments have shown that $\varepsilon \xi = 4.7$ gives the most satisfactory results.

3. Results and discussion

In order to validate the accuracy of Riemann-sum approximation method, we used MATLAB software to obtain numerical values of the dimensionless temperature of the surface using Eq. (8), and compared our results with those presented by Zanchini and Pulvirenti [4]. For numerical computation and comparison, we have considered the same numerical values for the controlling parameters as used by [4] and $M = 1000$. The numerical values for temperature at surface of the cylinder are obtained by Riemann-sum approximation and compared with [4] are presented in Table 1. It is seen to give an excellent agreement.

Table 1 Comparison of Benchmark values of θ at $\eta = 1$ for sinusoidal heat flux, $Fo = 5000$ and $\xi/Fo \leq 10$.

$\frac{\xi}{Fo}$	$\theta(\eta, \xi)$ (present work)	Ref. [4]
0.05	0.8341	0.8338
0.5	0.9162	0.9161
1.0	-0.6996	-0.6995
1.5	0.8278	0.8277
2.0	-0.7365	-0.7365
2.5	0.8075	0.8074
3.0	-0.7494	-0.7494
3.5	0.7986	0.7984
4.0	-0.7558	-0.7560
4.5	0.7937	0.7934
5.0	-0.7597	-0.7599
5.5	0.7905	0.7902
6.0	-0.7623	-0.7625
6.5	0.7884	0.7880
7.0	-0.7641	-0.7644
7.5	0.7868	0.7864
8.0	-0.7655	-0.7658
8.5	0.7857	0.7851
9.0	-0.7665	-0.7669
9.5	0.7847	0.7842
10.0	-0.7673	-0.7678

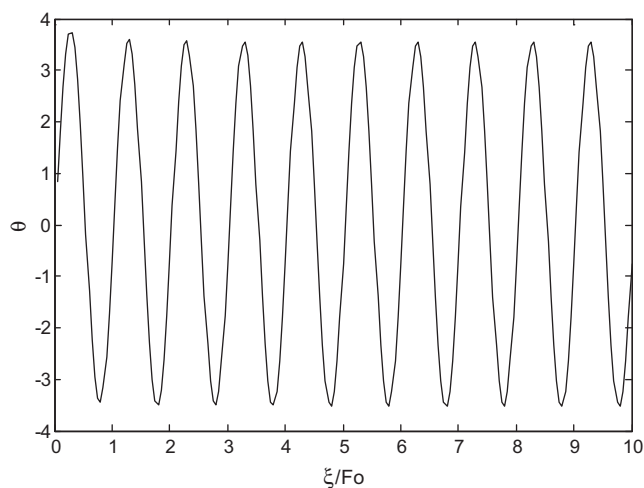
**Fig. 2** Temperature profile versus ξ/Fo for sinusoidal heat flux, with $Fo = 5000$, and $\eta = 1$.

Fig. 2 depicts dimensionless temperature θ at the surface of the cylinder ($\eta = 1$) for sinusoidal heat flux variation for $Fo = 5000$ and $0 \leq \xi/Fo \leq 10$. The graph is exactly the same as depicted in [4].

4. Conclusions

The Riemann-sum approximation approach has been employed to invert the Laplace domain to time domain for the temperature distribution around a cylindrical surface subjected to uniform heat flux sinusoidally varying with time. The numerical values of the surface temperature obtained by this method are in very good agreement with the benchmark values reported by Zanchini and Pulvirenti [4]. The results show that the Riemann-sum approximation approach can yield accurate values of the temperature field for the considered problem with less computational efforts.

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